Theoretical Equivalence and Invariant Structure

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- If T_1 and T_2 are equivalent, then what must they share in common? What are the "invariants" of theoretical equivalence?
- What is preserved in theory change? What are the "invariants" of theory change?
- What parts of our current theories ought we to believe?

Purported theoretical equivalences

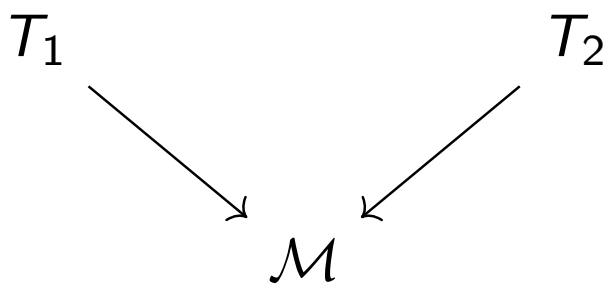
Schrödinger	$\cong$	Heisenberg	(von Neumann)
Lagrangian	$\cong$	Hamiltonian	(Legendre)
Newtonian Gravity	$\cong$	Newton-Cartan Gravity	(Weatherall)
Spacetimes	$\cong$	Leibniz Algebras	
Geometry of Points	$\cong$	Geometry of Lines	
Mereological Nihilism	$\cong$	Compositionalism	(Hirsch)
Relativism	$\cong$	Expressivism	(Dreier)

Naive accounts of equivalence

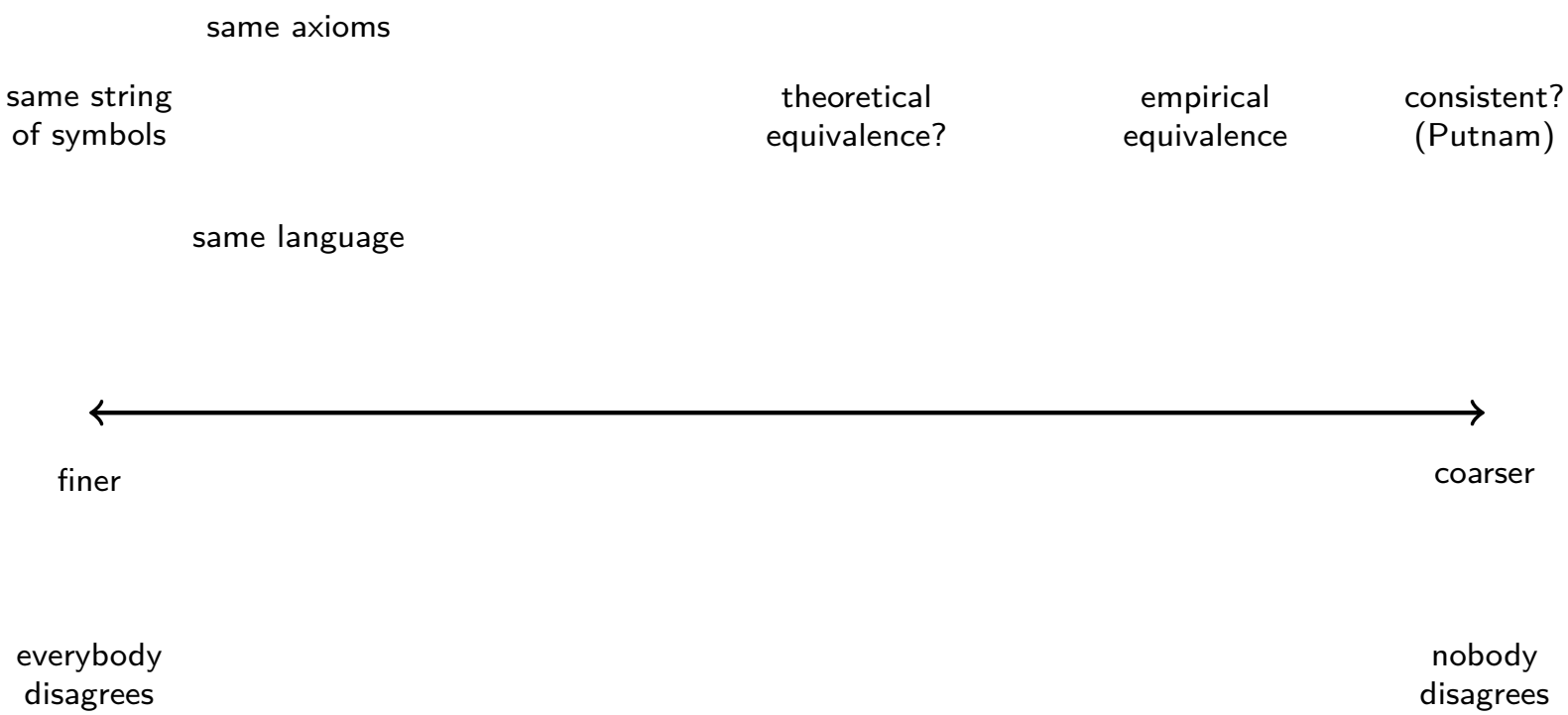
- T_1 and T_2 are equivalent just in case they say the same thing, i.e. they have the same content.
  - Correct but uninformative. How to specify content?
- T_1 and T_2 are equivalent just in case they make the same predictions.
- T_1 and T_2 are equivalent just in case they are true in exactly the same possible worlds.
- T_1 and T_2 are equivalent just in case they have the same set of models.

"Theories are extralinguistic entities which may be described or characterized by a number of different linguistic formulations." Suppe 1977

"While a theory may have many different formulations, its set of models is what is important." van Fraassen 2008



Spectrum of equivalence



Definitional equivalence

- Empirical equivalence was a suitable notion for verificationists, circa 1920–1950.
- The idea of definitionally equivalent or synonymous theories due to Tarski or Montague, circa 1960.
- Co-opted for philosophy of science by Glymour: "Theoretical realism and theoretical equivalence" 1970.

Definitional equivalence

	T_1	T_2
Language	$\{c\}$	$\{P\}$
Axioms	$\emptyset$	$\exists! x P x$

Definitional equivalence

	T_1	T_2
Language	$\{\circ, -1, e\}$	$\{\circ, e\}$
Axioms	associativity e is an identity $x \circ x^{-1} = e = x^{-1} \circ x$	associativity e is an identity $\exists! y(x \circ y = e = y \circ x)$

Interpretations and model maps

Definition

An interpretation or translation of T_1 into T_2 consists of a map F that takes n-place relation symbols of L_1 to n-place open formulas of L_2, and such that:

$$T_1 \vdash \phi \qquad \Longrightarrow \qquad T_2 \vdash \phi^F$$

$$\begin{array}{ccc} T_1 & & \mathcal{M}_1 \\ \Big\downarrow F & & \Big\uparrow F^* \\ T_2 & & \mathcal{M}_2 \end{array}$$

Definitional extension

Definition

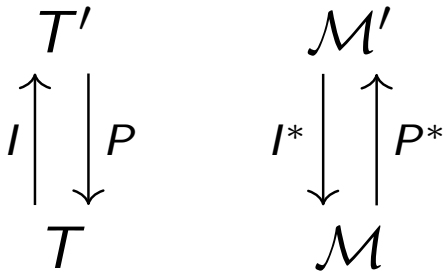
T' is a definitional extension of T if:

- T' explicitly defines each predicate symbol R of L' in terms of the language L:

$$T' \models \forall x(R(x) \leftrightarrow \phi_R(x))$$

- T' is a conservative extension of T.

Definitional extension



- T and T' are mutually interpretable.
- For each sentence ϕ of L'

$$T' \models (IP\phi) \leftrightarrow \phi$$

- For each sentence ψ of L

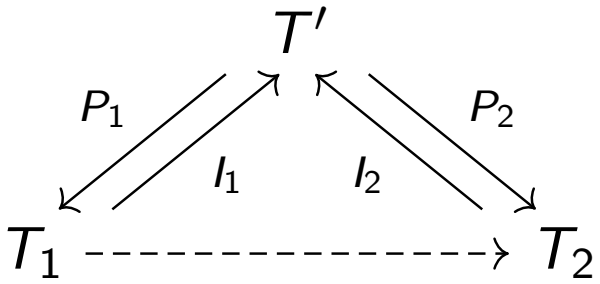
$$T \models (PI\psi) \leftrightarrow \psi$$

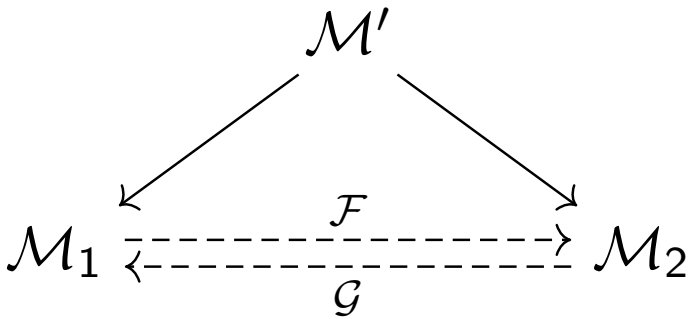
- The category of models of T is isomorphic to the category of models of T'.

Common definitional extension

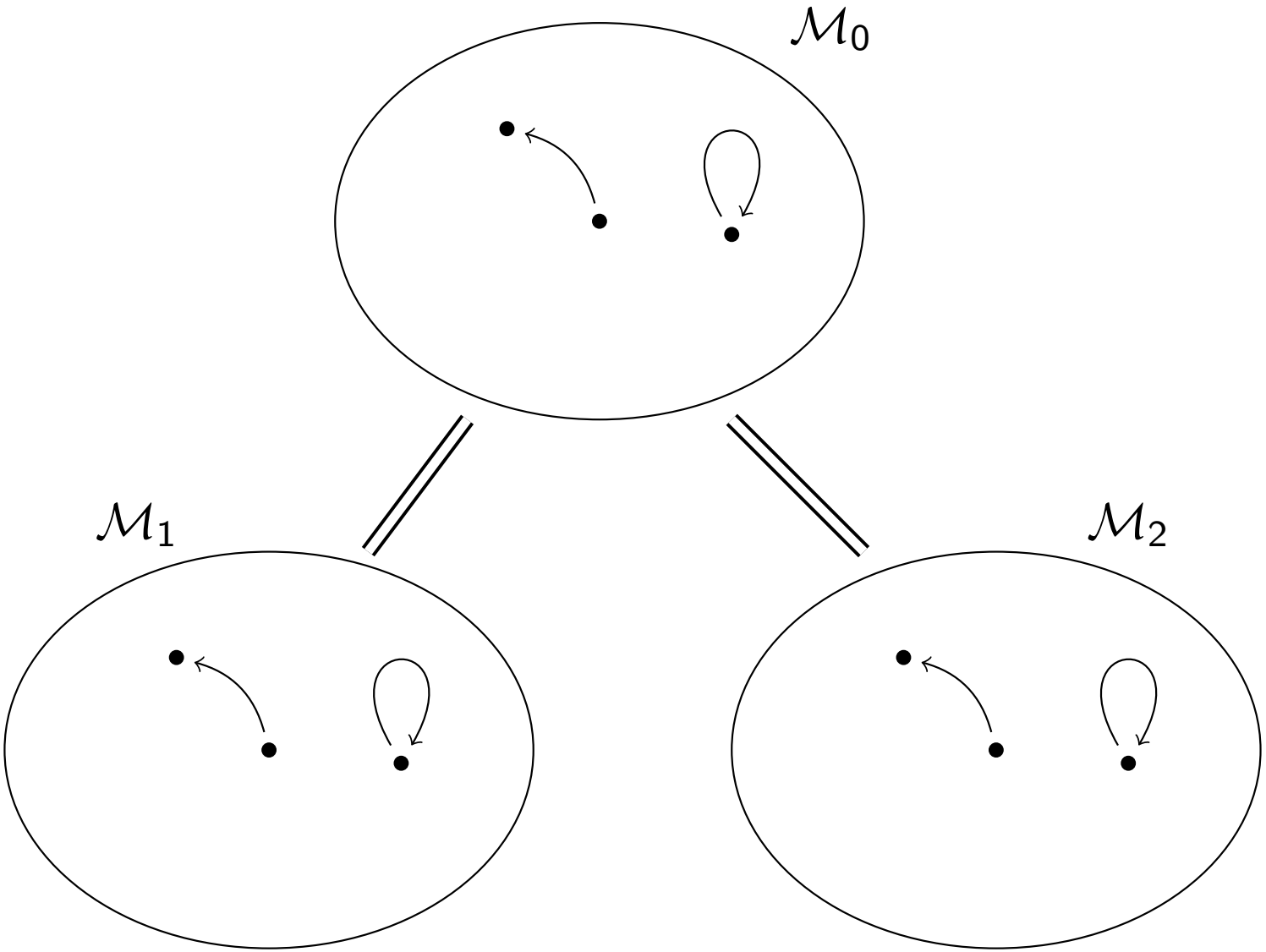
Definition

Two theories T_1 and T_2 are definitionally equivalent just in case they have a common definitional extension.

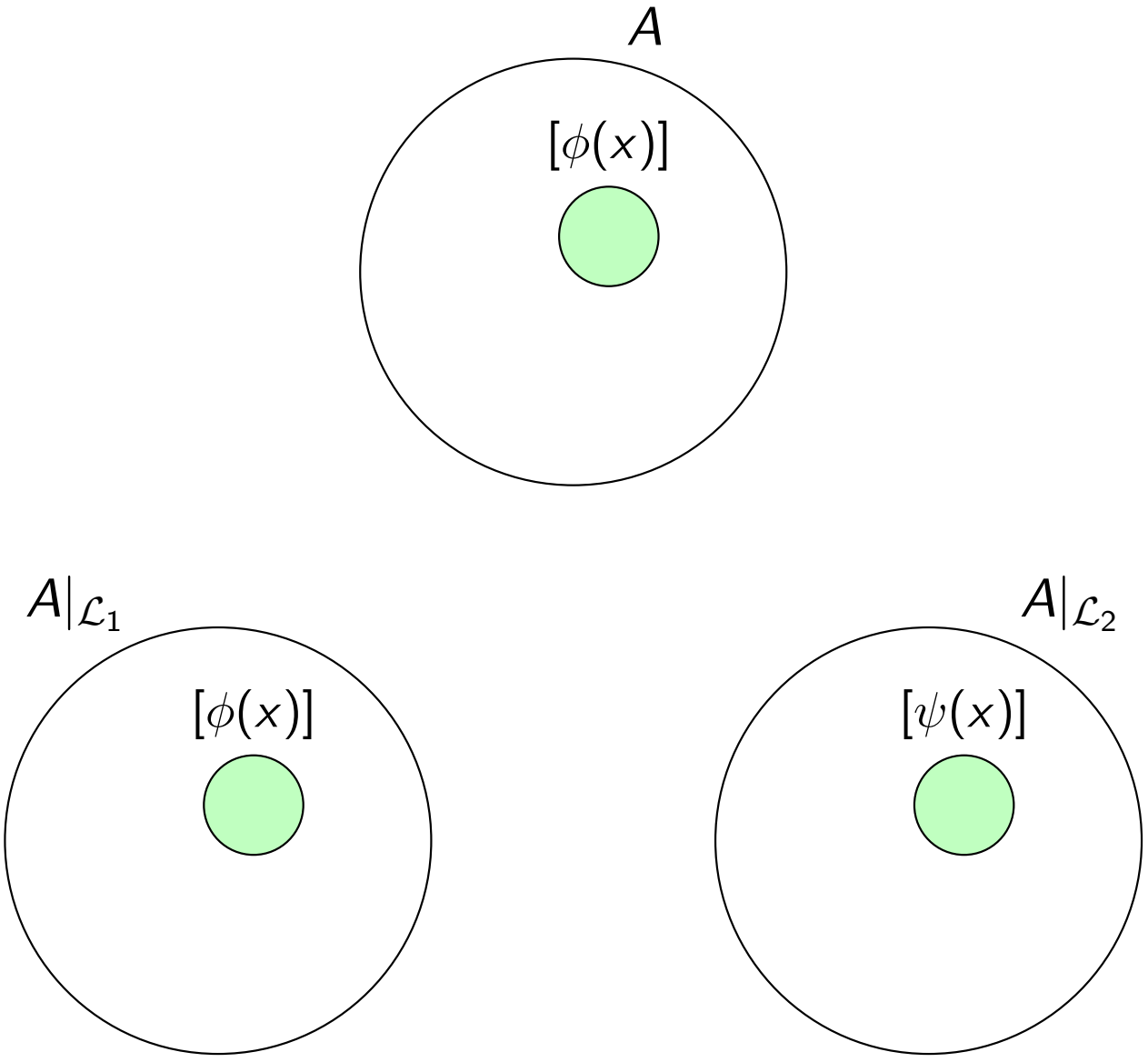




Invariants of definitional equivalence



These models describe the same world



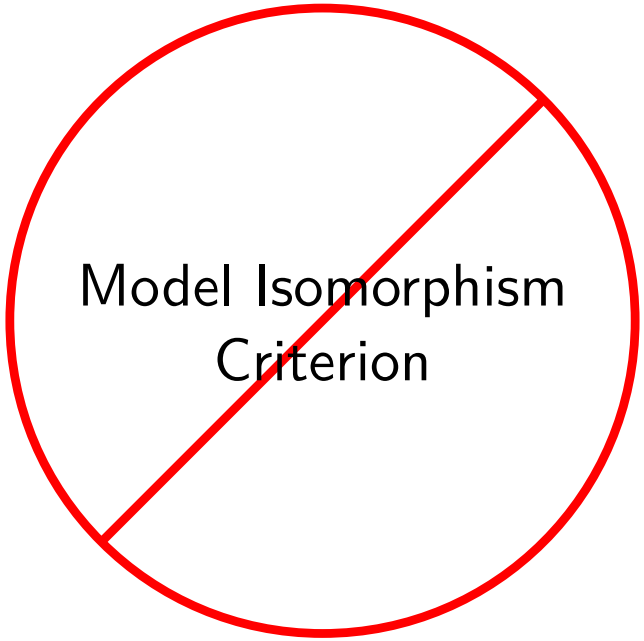
- $\mathcal{F} : \mathcal{M}_1 \to \mathcal{M}_2$ is an equivalence of categories.
- $\mathcal{F}(A)$ and A are *isomorphic*
  - have the same symmetry groups;
  - domains have same cardinality;
  - same definable sets.

Cutting nature at its joints

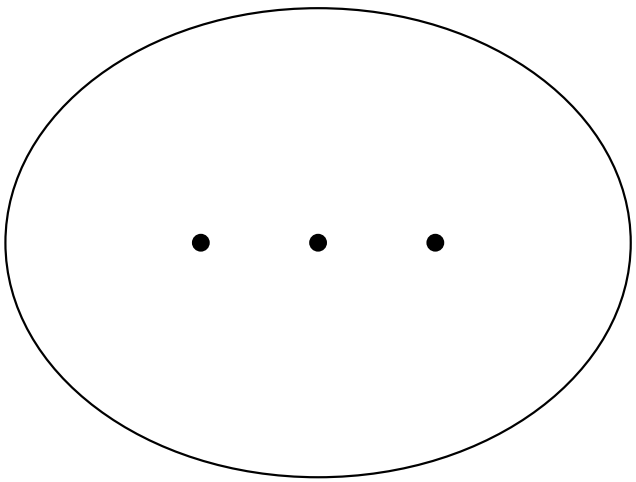
Model Isomorphism Criterion: If two theories are equivalent then their models are pairwise isomorphic.

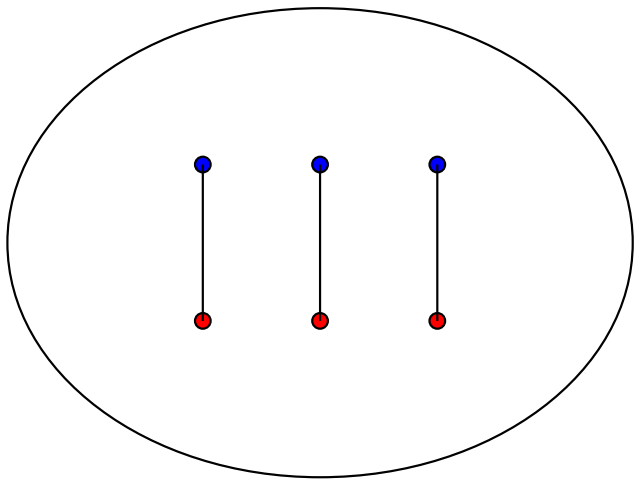
- Hamiltonian mechanics and Lagrangian mechanics are not equivalent.
- Spacetime theories and Leibniz algebra theories are not equivalent.

Structural Realism: A theory is true if the WORLD is isomorphic to one of its models.


Model Isomorphism
Criterion

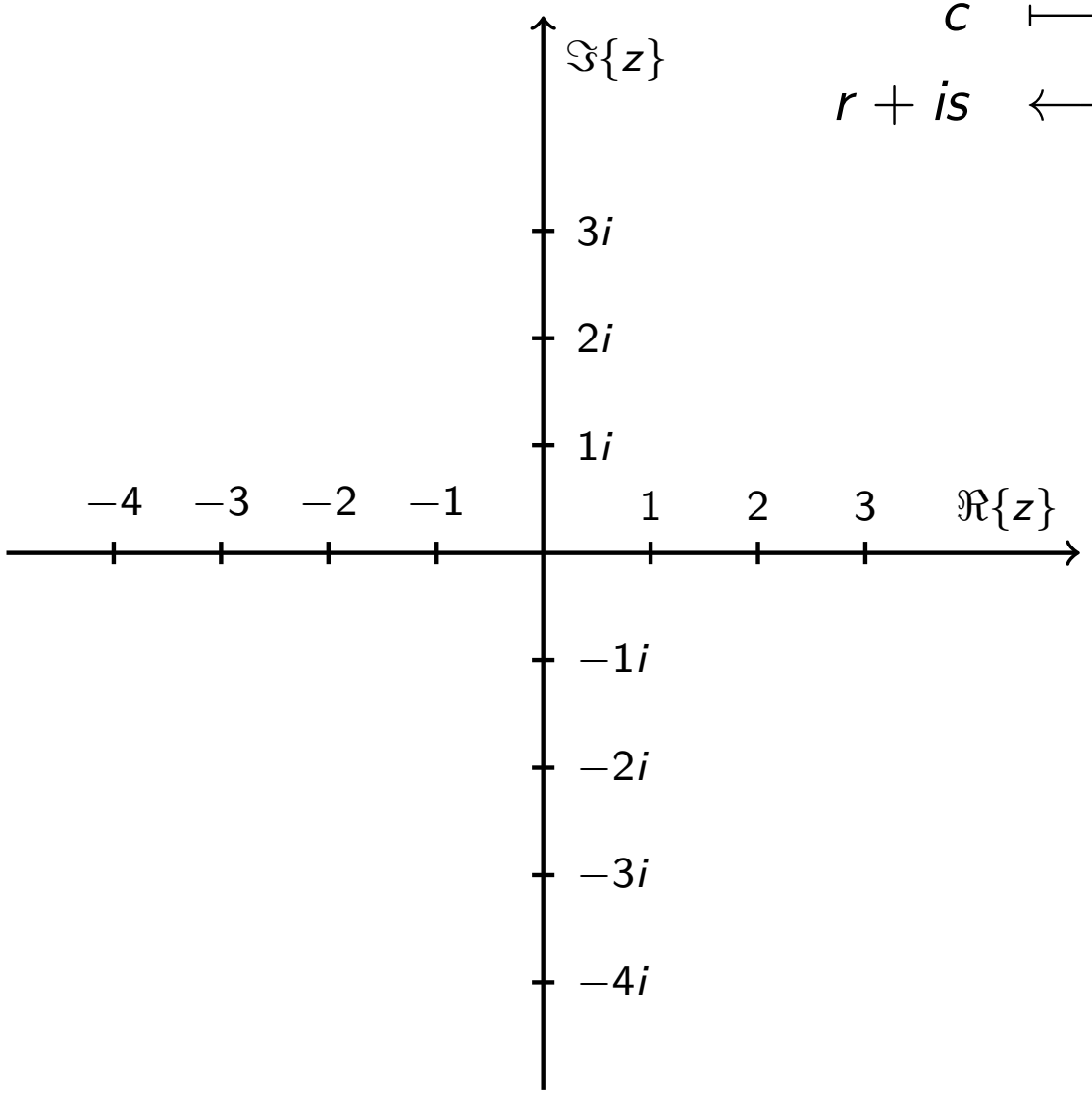
Against the model isomorphism criterion





Against the model isomorphism criterion

$$c \longmapsto \langle \mathrm{Re}(c), \mathrm{Im}(c) \rangle$$
$$r + is \longmapsto \langle r, s \rangle$$



Against the model isomorphism criterion

- A theory already implicitly quantifies over n-tuples:

$$\exists x_1 \exists x_2 \cdots \exists x_n \phi(x_1, \ldots, x_n)$$

- Tarski, Szczerba: we must recognize equivalence of geometries with *points* as primitives and geometries with *lines* as primitives

Multi-sorted logic

- Begin with a collection $A, B, C, \ldots$ of sorts.
- Each symbol (relations, functions, variables) of $\mathcal{L}$ is required to be coded by sorts. example $R : A, A, B$
- Quantifiers only range over particular sorts.
- Equality is sort-relative: $=_A$
- An $\mathcal{L}$ structure assigns a set/domain/universe to each sort.

Defining new universes

Product sorts (products): The formula $(x_1 = x_1) \wedge \cdots \wedge (x_n = x_n)$ defines a product sort $S_1 \times \cdots \times S_n$

Reduced sorts (subobjects): A formula $\phi(\overline{x})$ of L defines a reduction of the sort $S_1 \times \cdots \times S_n$.

Proposal: T^+ should be considered a definitional extension of T.

Defining new universes

Suppose that the formula $\phi(x, y)$ defines an equivalence relation in T. Let T^{eq} be the theory that adds a new sort E, function symbol $f : A; E$, and axioms

$$\forall x^A \forall y^A (\phi(x, y) \leftrightarrow f(x) = f(y))$$
$$\forall z^E \exists x^A (f(x) = z)$$

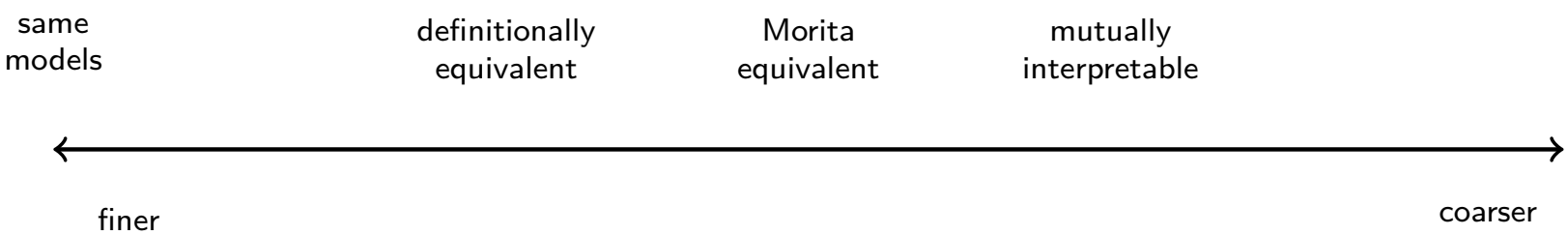
Proposal: T^{eq} should be considered a definitional extension of T.

Morita equivalence

The Hard Way: Each theory T has a classifying topos $\mathcal{E}[T]$. T_1 is Morita equivalent to T_2 just in case $\mathcal{E}[T_1]$ and $\mathcal{E}[T_2]$ are equivalent toposes.

The Easy Way: Theories T_1 and T_2 are Morita equivalent just in case they have a common definitional extension — in the extended sense where new sorts are definable from old.

Logical accounts of equivalence



Invariants

Morita equivalence: equivalent categories of models

Definitional equivalence: cardinalities of individual models, definable subsets

Note: "Equivalent categories of models" is necessary but not sufficient for theoretical equivalence.

Example: X versus $X \cup \{*\}$.

The case for Morita equivalence

- Argument by squeezing: Mathematical examples suggest that other criteria are too fine.
- Argument from mathematical convergence: It's a natural notion, both in logic and in category theory.
- Realizes Putnam's dream of sophisticated realism. (He himself abandoned hope!)

Concluding unscientific proposal

Proposal: If $T_1 \cong T_2$, then one should commit only to the invariants, i.e. features on which T_1 and T_2 agree.

Problem: The models of T_1 and T_2 might not agree on anything interesting.

Prospect: There is a more nuanced way in which the models describe the same world.